22. Rings of Polynomials

1. $f(x)+g(x)=2x^2+5, \quad f(x)g(x)=6x^2+4x+6$

2. $f(x)+g(x)=0, \quad f(x)g(x)=x^2+1$

3. $f(x)+g(x)=5x^2+5x+1, \quad f(x)g(x)=x^3+5x$

4. $f(x)+g(x)=3x^4+2x^3+4x^2+1,$
$f(x)g(x)=x^7+2x^6+4x^5+x^3+2x^2+x+3$

5. Such a polynomial is of the form ax^3+bx^2+cx+d where each of a,b,c,d may be either 0 or 1. Thus there are $2\cdot 2\cdot 2\cdot 2=16$ such polynomials in all.

6. Such a polynomial is of the form ax^2+bx+c where each of a,b,c maybe either 0, 1, 2, 3, or 4. Thus there are $5\cdot 5\cdot 5=125$ such polynomials in all.

7. $\phi_2(x^2+3)=2^2+3=7$

8. $\phi_i(2x^3-x^2+3x+2)=2i^3-i^2+3i+2=-2i+1+3i+2=i+3$

9. $\phi_3[(x^4+2x)(x^3-3x^2+3)]=\phi_3(x^4+2x)\phi_3(x^3-3x^2+3)=(3^4+6)(3^3-3^3+3)=(4+6)(3)=2$

10. $\phi_5[(x^3+2)(4x^2+3)(x^7+3x^2+1)]=\phi_5(x^3+2)\phi_5(4x^2+3)\phi_5(x^7+3x^2+1)=$
$(5^3+2)(4\cdot 5^2+3)(5^7+3\cdot 5^2+1)=(6+2)(2+3)(5+5+1)=(1)(5)(4)=6$

11. $\phi_4(3x^{106}+5x^{99}+2x^{53})=3(4)^{106}+5(4)^{99}+2(4)^{53}=3(4^6)^{17}4^4+5(4^6)^{16}4^3+2(4^6)^8 4^5$
$=3(1)4+5(1)1+2(1)2=5+5+4=0$

12. $1^2+1=0$ but $0^2+1\neq 0$, so 1 is the only zero.

13. Let $f(x)=x^3+2x+2$. Then $f(0)=2, f(1)=5, f(2)=0, f(3)=0, f(-3)=4, f(-2)=4$, and $f(-1)=6$ so 2 and 3 are the only zeros.

14. Let $f(x)=x^5+3x^3+x^2+2x$. Then $f(0)=0, f(1)=2, f(2)=4, f(-2)=4$, and $f(-1)=0$ so 0 and 4 are the only zeros.

15. Because $\mathbb{Z}_7$ is a field, $f(a)g(a)=0$ if and only if either $f(a)=0$ or $g(a)=0$. Let $f(x)=x^3+2x^2+5$ and $g(x)=3x^2+2x$. Then $f(0)=5, f(1)=1, f(2)=0, f(3)=1, f(-3)=3, f(-2)=5$, and $f(-1)=6$ while $g(0)=0, g(1)=5, g(2)=2, g(3)=5, g(-3)=0, g(-2)=1$, and $g(-1)=1$. Thus $f(x)g(x)$ has 0, 2, and 4 as its only zeros.

16. $\phi_3(x^{231}+3x^{117}-2x^{53}+1)=3^{231}+3^{118}-2(3^{53})+1=(3^4)^{57}3^3+(3^4)^{29}3^2-2(3^4)^{13}3+1=3^3+3^2-2(3)+1=2+4-1+1=1.$

17. Let $f(x)=2x^{219}+3x^{74}+2x^{57}+3x^{44}=2(x^4)^{54}x^3+3(x^4)^{18}x^2+2(x^4)^{14}x+3(x^4)^{11}$. Then $f(0)=0, f(1)=2+3+2+3=0, f(2)=1+2+4+3=0, f(-2)=4+2+1+3=0$ and $f(-1)=3+3+3+3=2$. Thus 0, 1, 2,and 3 are zeros of $f(x)$.

18. The definition is incorrect. All but a finite number of the a_i must be zero.

A **polynomial with coefficients in a ring** R is an infinite formal sum

$$\sum_{i=1}^{\infty} a_i x^i = a_0+a_1x+a_2x^2+\cdots+a_nx^n+\cdots$$

where $a_i\in R$ for $i=0,1,2,\cdots$ and all but a finite number of the a_i are 0.

19. The definition is incorrect. The zero α may be in a field E containing F.

Let F be a subfield of a field E and let $f(x) \in F[x]$. A **zero of** $f(x)$ **in** E is an $\alpha \in E$ such that $\phi_\alpha(f(x)) = 0$, where $\phi_\alpha : F[x] \to E$ is the evaluation homomorphism mapping x into α.

20. $f(x,y) = (3x^3 + 2x)y^3 + (x^2 - 6x + 1)y^2 + (x^4 - 2x)y + (x^4 - 3x^2 + 2) =$
$(y+1)x^4 + 3y^3x^3 + (y^2 - 3)x^2 + (2y^3 - 6y^2 - 2y)x + (y^2 + 2)$

21. (See the answer in the text.)

22. $2x + 1$ is a unit because $(2x+1)^2 = 1$ in $\mathbb{Z}_4[x]$.

23. T T T T F F T T T F

24. Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ and $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$ be polynomials in $D[x]$ with a_n and b_m both nonzero. Because D is an integral domain, we know that $a_n b_m \neq 0$, so $f(x)g(x)$ is nonzero because its term of highest degree has coefficient $a_n b_m$. As stated in the text, $D[x]$ is a commutative ring with unity, and we have shown it has no divisors of zero, so it is an integral domain.

25. **a.** The units in $D[x]$ are the units in D because a polnomial of degree n times a polynomial of degree m is a polynomial of degree nm, as proved in the preceding exercise. Thus a polynomial of degree 1 cannot be multiplied by anything in $D[x]$ to give 1, which is a polynomial of degree 0.

b. They are the units in $\mathbb{Z}$, namely 1 and -1.

c. They are the units in $\mathbb{Z}_7$, namely 1, 2, 3, 4, 5, and 6.

26. Let $f(x) = \sum_{i=0}^{\infty} a_i x^i, g(x) = \sum_{i=0}^{\infty} b_i x^i$, and $h(x) = \sum_{i=0}^{\infty} c_i x^i$. Then

$$\begin{aligned}
h(x)[f(x)+g(x)] &= \left[\sum_{j=0}^{\infty} c_j x^j\right]\left[\sum_{i=0}^{\infty}(a_i+b_i)x^i\right] = \sum_{n=0}^{\infty}\left[\sum_{i=0}^{n} c_i(a_{n-i}+b_{n-i})\right]x^n \\
&= \sum_{n=0}^{\infty}\left[\sum_{i=0}^{n} c_i a_{n-i}\right]x^n + \sum_{n=0}^{\infty}\left[\sum_{i=0}^{n} c_i b_{n-i}\right]x^n \\
&= \left[\sum_{j=0}^{\infty} c_j x^j\right]\left[\sum_{i=0}^{\infty} a_i x^i\right] + \left[\sum_{j=0}^{\infty} c_j x^j\right]\left[\sum_{i=0}^{\infty} b_i x^i\right] \\
&= h(x)f(x) + h(x)g(x)
\end{aligned}$$

so the left distributive law holds.

27. **a.** Let $f(x)\sum_{i=0}^{\infty} a_x^i$ and $g(x) = \sum_{i=0}^{\infty}$ be polynomials in $F[x]$. Then

$$\begin{aligned}
D(f(x)+g(x)) &= D\left(\sum_{i=0}^{\infty} a_i x^i + \sum_{i=0}^{\infty} b_i x^i\right) = D\left(\sum_{i=0}^{\infty}(a_i+b_i)x^i\right) \\
&= \sum_{i=1}^{\infty} i(a_i+b_i)x^{i-1} = \sum_{i=1}^{\infty}(ia_i + ib_i)x^{i-1} \\
&= \sum_{i=1}^{\infty} i a_i x^{i-1} + \sum_{i=1}^{\infty} i b_i x^{i-1} = D(f(x)) + D(g(x)).
\end{aligned}$$

b. The kernel of D is F. [This would not be true if F had characteristic p, for then $D(x^p) = 0$.]

c. The image of D is $F[x]$ because D is additively a homomorphism with $D(1) = 0$ and $D(\frac{1}{i+1}a_i x^{i+1}) = a_i x^i$.

28. a. $\phi_{\alpha_1,\cdots,\alpha_n}(f(x_1,\cdots,x_n))$ is the element of F obtained by replacing each x_i by α_i in the polynomial and computing in E the resulting sum of products. That is $\phi_{\alpha_1,\cdots,\alpha_n}(f(x_1,\cdots,x_n)) = f(\alpha_1,\cdots,\alpha_n)$. This is a map $\phi_{\alpha_1,\cdots,\alpha_n} : F[x_1,\cdots,x_n] \to E$ which is a homomorphism and maps F isomorphically by the identity map, that is, $\phi_{\alpha_1,\cdots,\alpha_n}(a) = a$ for $a \in F$.

b. $\phi_{-3,2}(x_1^2x_2^3 + 3x_1^4x_2) = (9)(8) + 3(81)(2) = 72 + 486 = 558$.

c. Let F be a subfield of E. Then $(\alpha_1,\cdots,\alpha_n)$ in $\underbrace{E \times E \times \cdots E}_{n \text{ factors}}$ is a **zero** of $f(x_1,\cdots,x_n) \in F[x_1,\cdots x_n]$ if $\phi_{\alpha_1,\cdots,\alpha_n}(f(x_1,\cdots,x_n)) = 0$.

29. *Addition associative:* Let $\phi,\psi,\mu \in R^R$. Then $[(\phi+\psi)+\mu](r) = (\phi+\psi)(r)+\mu(r) = \phi(r)+\psi(r)+\mu(r) = \phi(r) + (\psi+\mu)(r) = [\phi + (\psi+\mu)](r)$ because addition in R is associative. Because $(\phi+\psi)+\mu$ and $\phi+(\psi+\mu)$ have the same value on each $r \in R$, they are the same function.

Identity for $+$: The function ϕ_0 such that $\phi_0(r) = 0$ for all $r \in R$ acts as additive identity, for $(\phi_0+\psi)(r) = \phi_0(r)+\psi(r) = 0+\psi(r) = \psi(r)$. Because ϕ_0 and $\phi_0+\psi$ have the same value on each $r \in R$, we see that they are the same function. A similar argument shows that $\psi+\phi_0 = \psi$.

Additive inverse: Given $\phi \in R^R$, the function $-\phi$ defined by $(-\phi)(r) = -(\phi(r))$ for $r \in R$ is the additive inverse of ϕ, for $(\phi+(-\phi))(r) = \phi(r)+(-\phi)(r) = \phi(r)+(-\phi(r)) = 0 = \phi_0(r)$, so $\phi+(-\phi) = \phi_0$. A similar argument shows that $(-\phi)+\phi = \phi_0$.

Addition commutative: We have $(\phi+\psi)(r) = \phi(r)+\psi(r) = \psi(r)+\phi(r) = (\psi+\phi)(r)$ because addition in R is commutative. Thus $\phi+\psi$ and $\psi+\phi$ are the same function, so $\phi+\psi = \psi+\phi$.

Multiplication associative: Now $[(\phi\cdot\psi)\cdot\mu](r) = [(\phi\cdot\psi)(r)]\mu(r) = [\phi(r)\psi(r)]\mu(r) = \phi(r)[\psi(r)\mu(r)] = \phi(r)[(\psi\cdot\mu)(r)] = [\phi\cdot(\psi\cdot\mu)](r)$ because multiplication in R is associative. Thus $\phi\cdot(\psi\cdot\mu) = (\phi\cdot\psi)\cdot\mu$ because the functions have the same value on each $r \in R$.

Left distributive law: We have $[\phi\cdot(\psi+\mu)](r) = \phi(r)[(\psi+\mu)(r)] = \phi(r)[\psi(r)+\mu(r)] = \phi(r)\psi(r) + \phi(r)\mu(r) = (\phi\cdot\psi)(r) + (\phi\cdot\mu)(r) = [\phi\cdot\psi+\phi\cdot\mu](r)$ because the left distributive law holds in R. Thus $\phi\cdot(\psi+\mu) = \phi\cdot\psi+\phi\cdot\mu$ because these functions have the same value at each $r \in R$.

Right distributive law: The proof is analogous to that for the left distributive law.

30. a. The map $\mu : F[x] \to F^F$ where $\mu(f(x))$ is the function $\phi \in F^F$ such that $\phi(a) = f(a)$ for all $a \in F$ is easily seen to be a homomorphism of $F[x]$ into the ring F^F, and by definition, $P_F = \mu[F[x]]$. Thus P_F is the homomorphic image of a ring under a ring homomorphism. Theorem 13.12 then shows that $\langle P_F, +\rangle$ is a group. Let $\phi,\psi \in P_F$, and let $f(x), g(x) \in F[x]$ be such that $\mu(f(x)) = \phi$ and $\mu(g(x)) = \psi$. Then $(\phi\cdot\psi)(a) = \phi(a)\psi(a) = f(a)g(a)$ for all $a \in F$, so $\mu(f(x)g(x)) = \phi\cdot\psi$. Thus $\phi\cdot\psi \in P_F$ so P_F is closed under multiplication. By Exercise 48 of Section 18, P_F is a subring of F_F.

b. Let F be the finite field $\mathbb{Z}_2$. A function in $\mathbb{Z}_2^{\mathbb{Z}_2}$ has just two elements in both its domain and range. Thus there are only $2^2 = 4$ such functions in all. However, $\mathbb{Z}_2[x]$ is an infinite set, so it isn't isomorphic to $P_{\mathbb{Z}_2}$.

31. a. There are $2^2 = 4$ elements in $\mathbb{Z}_2^{\mathbb{Z}_2}$ and $3^3 = 27$ in $\mathbb{Z}_3^{\mathbb{Z}_3}$.

b. Because $(\phi+\phi)(a) = \phi(a)+\phi(a) = 2\cdot\phi(a) = 0$ in $\mathbb{Z}_2$, we see that every element of $\mathbb{Z}_2^{\mathbb{Z}_2}$ is its own additive inverse, so this additive group of order 4 must be isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. Similarly, if